

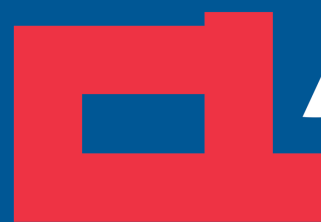
ASSOCIATION FOR COMPUTATIONAL LINGUISTICS (ACL 2026)

Rethinking **Jailbreak Detection** of Large Vision-Language Models with **Representational Contrastive Scoring**

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SAN DIEGO

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THE PROBLEM

What a deployable defense actually needs

LVLMs inherit text jailbreak — and add brand-new visual attack surface on top.

01

Generalizable

Catch attack types that
were never seen

02

Lightweight

Run in real time.

03

Low over-rejection

Especially for unseen benign
inputs

Today's defenses make the same trade-off

01

Safety alignment

Retrain the model (SFT / DPO / RLHF).

Expensive data + compute
broken by new attacks
can erode benign behavior.

02

Filters & guard LLMs

Bolt on an external judge or classifier.

Heavy and high-latency
tuned to known attacks
poor generalization.

03

← **our family**

Detect on model internals

Read the safety signal already inside.

Lightweight
no retraining
decides before generation

THE GAP

Anomaly detection confuses the **unseen** with **malice**

JailDAM (COLM 2025): Autoencoders on CLIP features

Anomaly detectors model only benign inputs and **flag whatever looks unusual**.

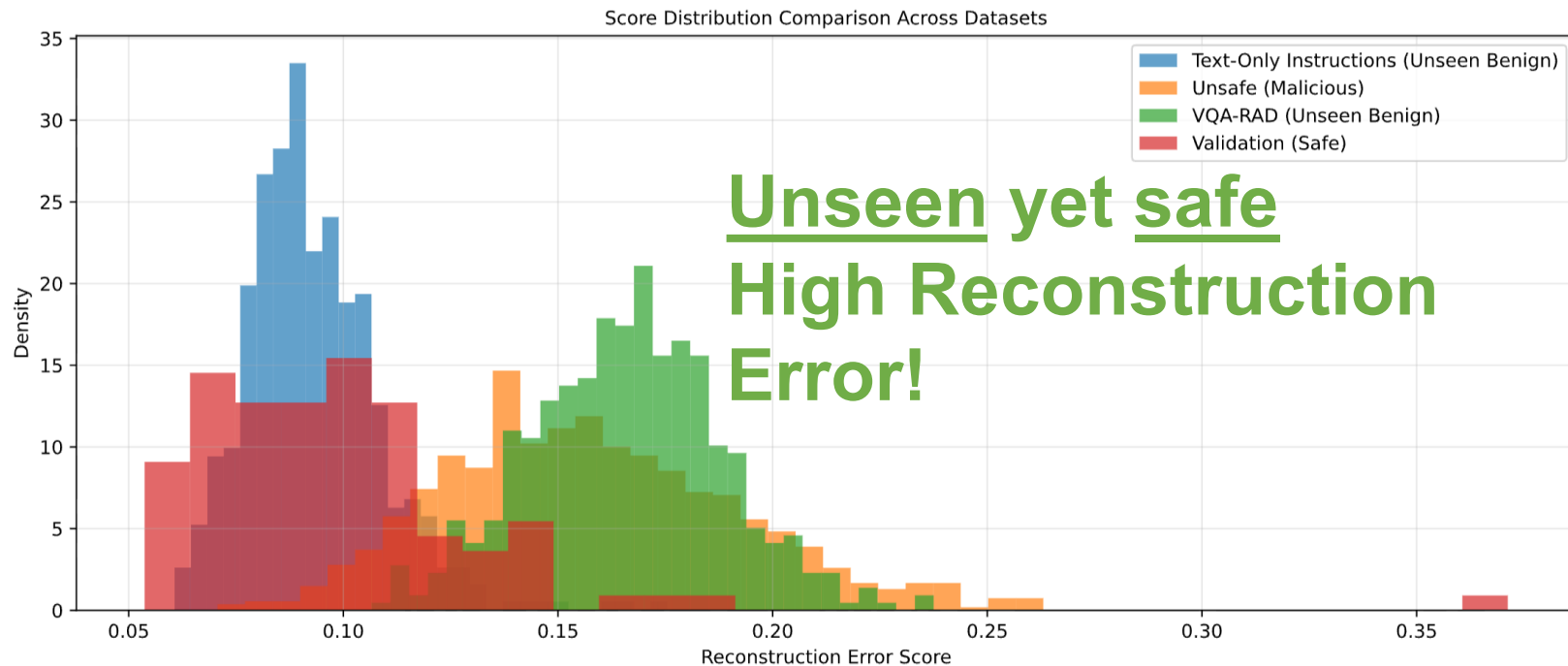
An **unseen-but-benign** input — e.g. medical images — looks just as anomalous as an attack.

0.91 → 0.71

AUROC, once unseen-benign data appears

95% → 57%

Severe over-rejection

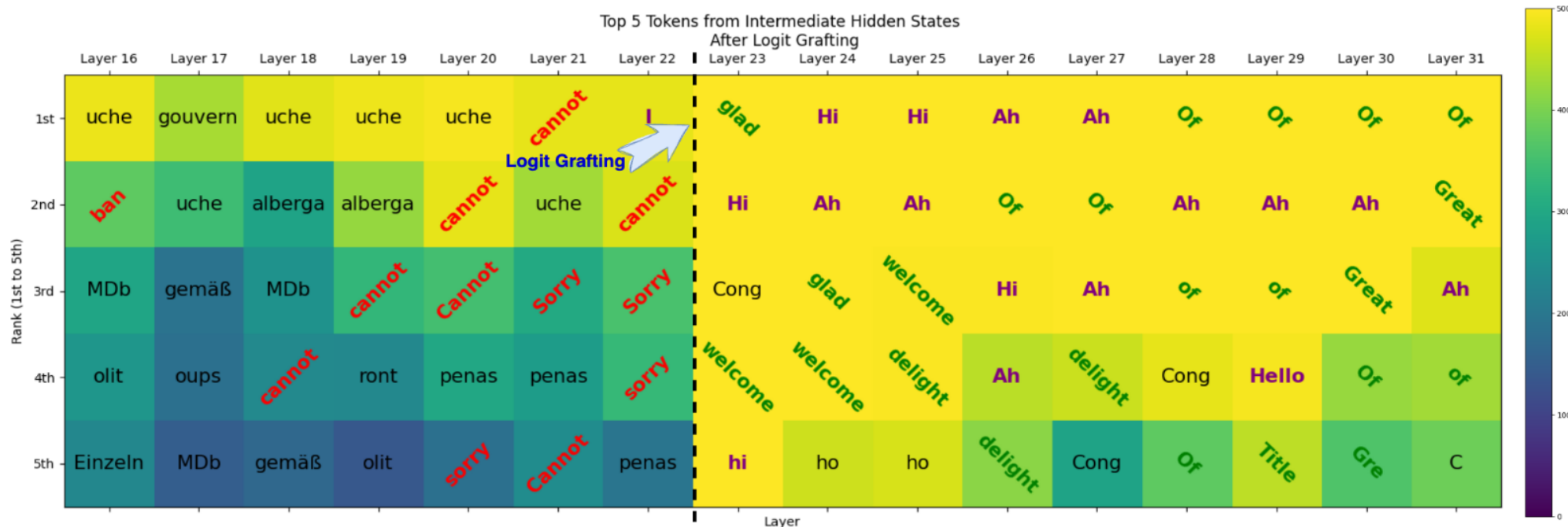


OUR IDEA

Read the model's own mind — contrastively

1 The signal is already inside under Logit Lens

LVLm's own intermediate representations encode intent better



Zhenhong Zhou, et al.
Findings of EMNLP
2024

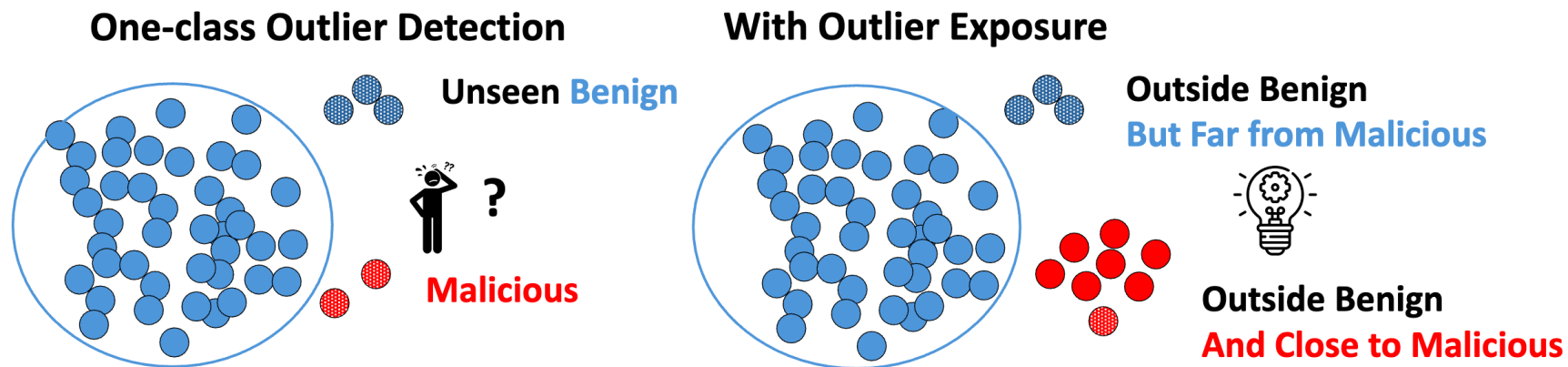
OUR IDEA

Read the model's own mind — **contrastively**

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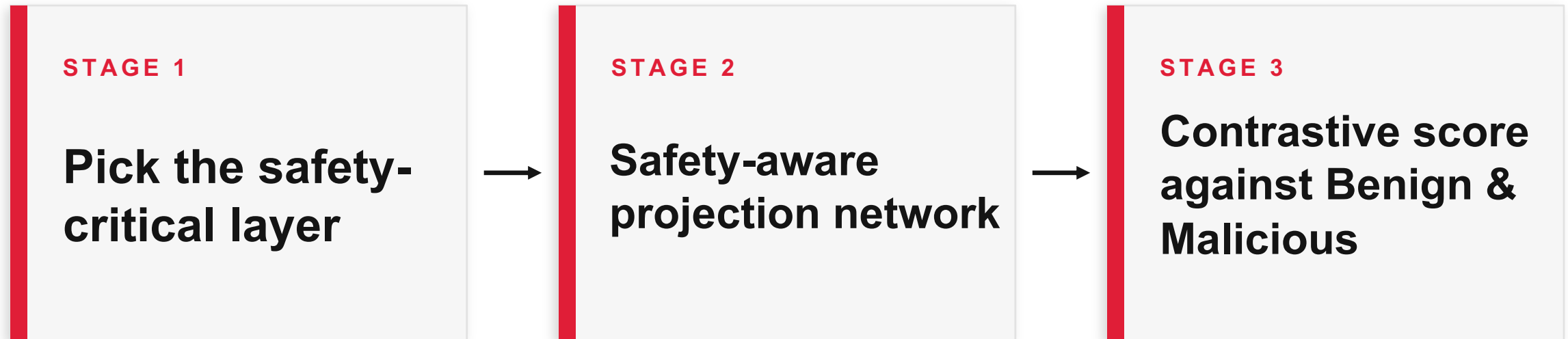
2 Use benign AND malicious

- **Outlier exposure** enhances anomaly detection
- Instead of “is this unusual?”, ask “closer to benign, or to malicious?” — a relative and easier decision.



FRAMEWORK

The pipeline — three stages



Decision before the first token — block jailbreaks proactively and skip generation cost on rejected prompts.

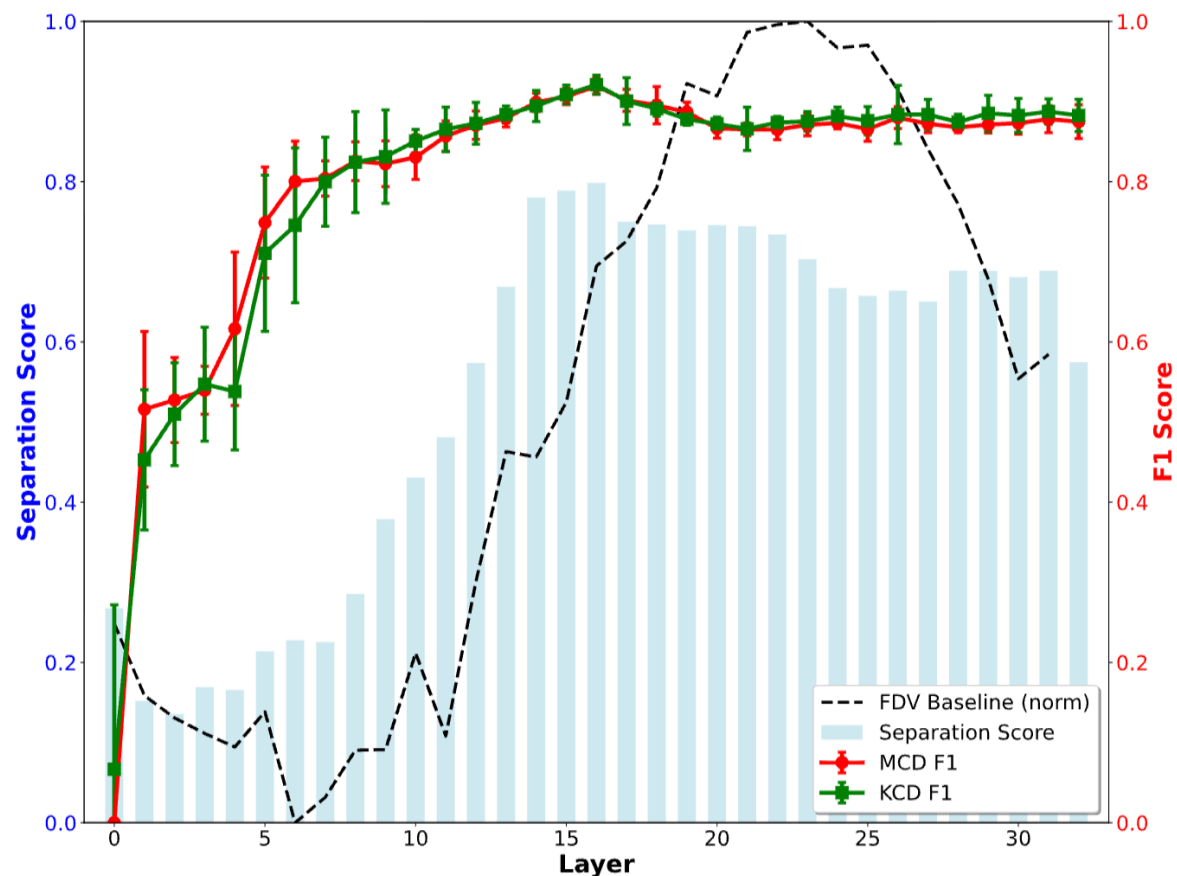
STAGE 1

Where does safety live in the network?

Score by **geometric separability** of benign vs. malicious prompts:

- SVM margin
- Silhouette score
- Inter- / intra-class distance ratio

Middle Layer is the Sweet Spot!



(b) F1 scores on LLaVA.

STAGES 2-3

Project the signal, then score by distance

Project (4096 $\rightarrow$ 256)

A lightweight MLP

Down-project while pushing benign and malicious apart.

Score: KCD & MCD

KCD uses nearest-neighbour distance;

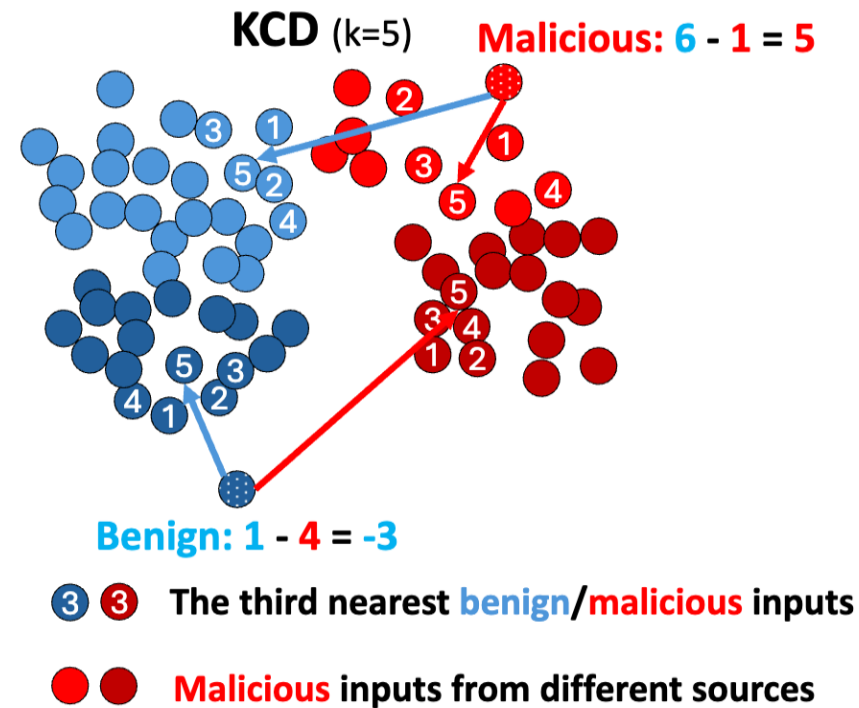
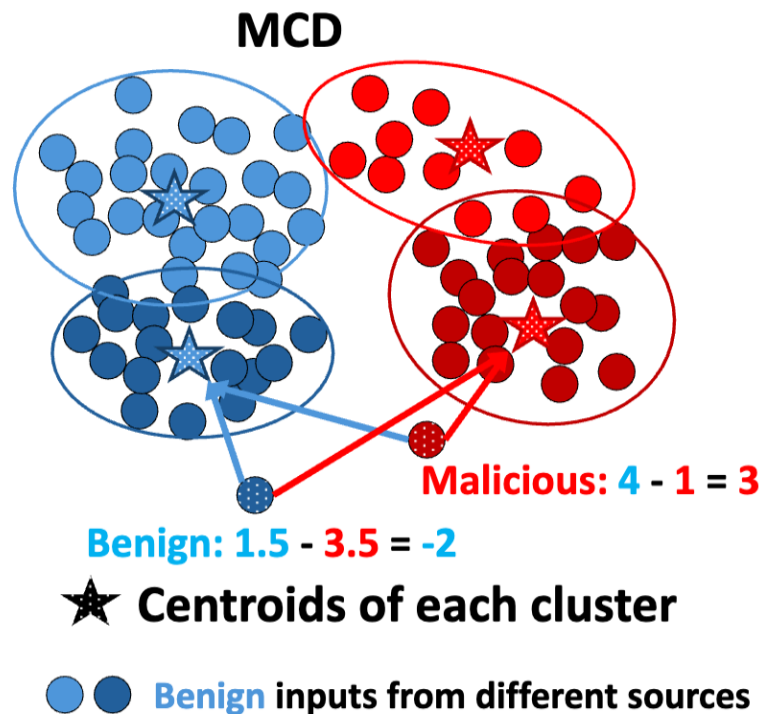
MCD models each group as a Gaussian and uses Mahalanobis distance.

STAGES 2-3

Project the signal, then score by distance

Project (4096 $\rightarrow$ 256)

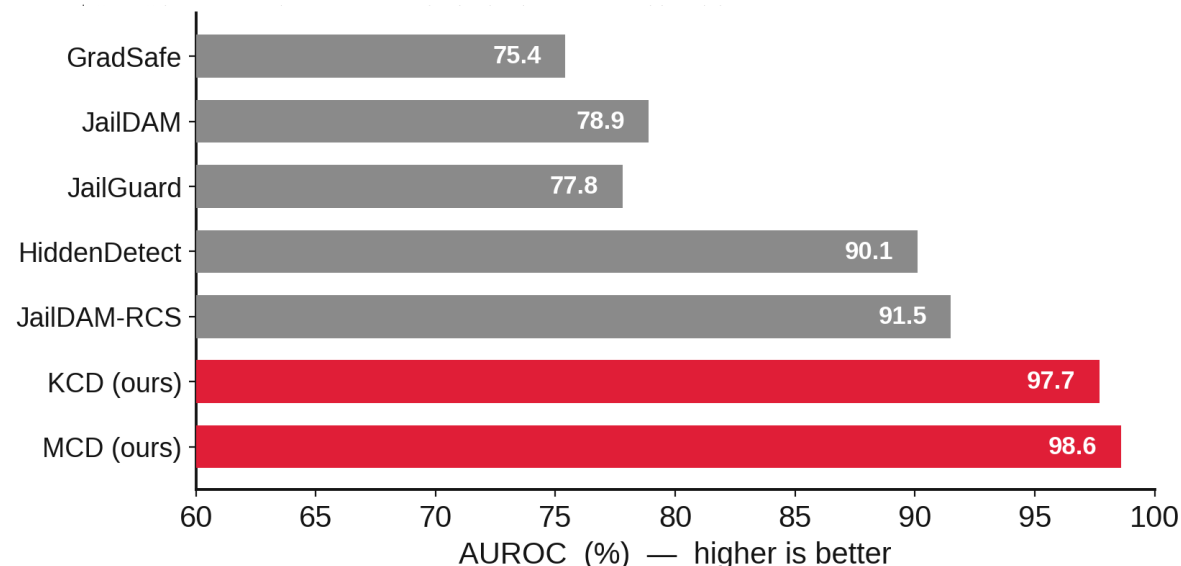
Score: KCD & MCD



RESULTS

State-of-the-art — on attacks **held out** from training

Split	Class	Source	Modality	# Samples
Training	Benign	Alpaca (Taori et al., 2023)	Text	500
		MM-Vet (Yu et al., 2024)	MM*	218
		OpenAssistant (Köpf et al., 2023)	Text	282
	Malicious	AdvBench (Zou et al., 2023b)	Text	300
		DAN (Shen et al., 2024)	Text	150
		JailbreakV-28K* (Luo et al., 2024)	MM	550
Testing	Benign	XSTest (Röttger et al., 2024)	Text	250
		FigTxt (Jiang et al., 2025a)	Text	300
		VizWiz (Gurari et al., 2018)	MM	450
	Malicious	XSTest (Röttger et al., 2024)	Text	200
		FigTxt (Jiang et al., 2025a)	Text	350
		VAE (Qi et al., 2024a)	MM	200
		JailbreakV-28K† (Luo et al., 2024)	MM	150



97.7 / 98.6

AUROC (LLaVA-7B)

+7 points over the best prior detector.

Trained on two attack families, tested on a **third, unseen** — a true generalization test.

>90% F1 Score

PRACTICALITY

Cheap to run, quick to adapt

$\leq 5.5\%$

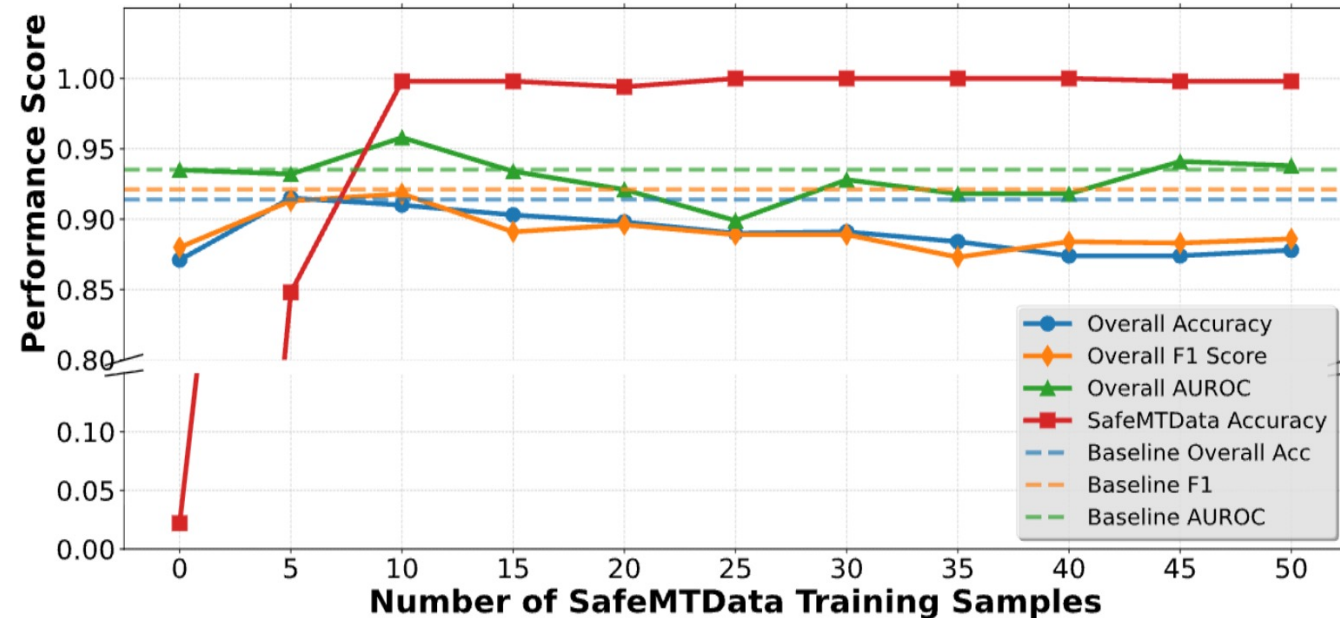
inference overhead vs. the host LVLM

< 0.015 GB memory

Decides before the first token.

Rejected prompts **never** pay the cost of generation.

KCD Performance vs SafeMTData Training Size (LLAVA)



Few-shot adaptation on more tricky cases

Future

Limitation & Future Work

- Agentic AI Jailbreak
- Multi-turn Jailbreak
- Interactive Evaluation
- Defense-in-depth Evaluation/Benchmark

TAKEAWAYS

Effective safety needs no retraining and no heavy external models.

Thank you!

Presenter:

Peichun Hua (peichunhua04@gmail.com, peichun.xyz)

Code: github.com/sarendis56/Jailbreak_Detection_RCS

Ad: I am actively looking for PhD opportunities in the cycle of Spring/Fall 2027 on AI security, safety, and privacy! Please let me know if you know such positions.